

EXP-5

COMPARATIVE ANALYSIS OF DIFFERENT TYPES OF PROMPTING PATTERNS AND EXPLAIN WITH VARIOUS TEST SCENARIOS

AIM: TO TEST AND COMPARE HOW DIFFERENT PATTERN MODELS RESPOND TO VARIOUS PROMPTS (BROAD OR UNSTRUCTURED) VERSUS BASIC PROMPTS (CLEARER AND MORE REFINED) ACROSS MULTIPLE SCENARIOS. ANALYZE THE QUALITY, ACCURACY, AND DEPTH OF THE GENERATED RESPONSES.

1.GENERATING A CREATIVE STORY:

BROAD / UNSTRUCTURED PROMPT:

PROMPT:

“WRITE A STORY.”

OUTPUT:

ONCE THERE WAS A BOY WHO FOUND A MYSTERIOUS KEY IN THE FOREST. HE SEARCHED FOR THE LOCK THAT MATCHED IT AND DISCOVERED AN OLD HOUSE. INSIDE, HE FOUND SECRETS ABOUT HIS FAMILY AND BEGAN AN ADVENTURE THAT CHANGED HIS LIFE FOREVER.

BASIC / REFINED PROMPT:

PROMPT:

“WRITE A SHORT FANTASY STORY ABOUT A 15-YEAR-OLD GIRL WHO DISCOVERS A MAGICAL BOOK IN HER SCHOOL LIBRARY.”

OUTPUT:

MAYA ALWAYS STAYED LATE IN THE SCHOOL LIBRARY BECAUSE SHE LOVED READING OLD BOOKS. ONE RAINY EVENING, SHE NOTICED A GLOWING BOOK HIDDEN BEHIND

A SHELF. WHEN SHE OPENED IT, GOLDEN LETTERS FLOATED IN THE AIR AND FORMED A MAP. THE MAP LED HER TO A HIDDEN ROOM BENEATH THE LIBRARY WHERE ANCIENT MAGICIANS ONCE TRAINED STUDENTS. TO PROTECT THE MAGICAL WORLD FROM DARKNESS, MAYA HAD TO SOLVE THREE RIDDLES BEFORE SUNRISE.

2. ANSWERING A FACTUAL QUESTION:

BROAD / UNSTRUCTURED PROMPT:

PROMPT:

“TELL ME ABOUT COMPUTERS.”

OUTPUT:

COMPUTERS ARE ELECTRONIC DEVICES USED TO PROCESS DATA. THEY ARE USED IN HOMES, SCHOOLS, BUSINESSES, AND MANY OTHER PLACES.

BASIC / REFINED PROMPT:

PROMPT:

“EXPLAIN THE DIFFERENCE BETWEEN RAM AND ROM IN SIMPLE TERMS WITH EXAMPLES.”

OUTPUT:

RAM (RANDOM ACCESS MEMORY) IS TEMPORARY MEMORY USED WHILE THE COMPUTER IS RUNNING. FOR EXAMPLE, WHEN YOU OPEN A BROWSER OR GAME, IT USES RAM. DATA IN RAM IS ERASED WHEN THE COMPUTER IS TURNED OFF.

ROM (READ ONLY MEMORY) IS PERMANENT MEMORY THAT STORES IMPORTANT STARTUP INSTRUCTIONS. FOR EXAMPLE, THE BIOS PROGRAM STORED IN ROM HELPS THE COMPUTER START PROPERLY.

3. SUMMARIZING AN ARTICLE OR CONCEPT:

BROAD / UNSTRUCTURED PROMPT:

PROMPT:

“SUMMARIZE CLIMATE CHANGE.”

OUTPUT:

CLIMATE CHANGE REFERS TO LONG-TERM CHANGES IN EARTH’S TEMPERATURE AND WEATHER PATTERNS CAUSED MAINLY BY HUMAN ACTIVITIES.

BASIC / REFINED PROMPT:

PROMPT:

“SUMMARIZE THE CAUSES AND EFFECTS OF CLIMATE CHANGE IN 5 BULLET POINTS FOR SCHOOL STUDENTS.”

OUTPUT

- BURNING FOSSIL FUELS INCREASES GREENHOUSE GASES.
- DEFORESTATION REDUCES CARBON ABSORPTION BY TREES.
- GLOBAL TEMPERATURES ARE RISING.
- ICE CAPS ARE MELTING AND SEA LEVELS ARE INCREASING.
- EXTREME WEATHER EVENTS LIKE FLOODS AND HEATWAVES ARE BECOMING MORE COMMON.

4. PROVIDING ADVICE OR RECOMMENDATIONS:

BROAD / UNSTRUCTURED PROMPT:

PROMPT:

“RECOMMEND A HOBBY.”

OUTPUT:

YOU CAN TRY READING, PAINTING, GARDENING, OR PLAYING SPORTS.

BASIC / REFINED PROMPT

PROMPT:

“SUGGEST LOW-COST HOBBIES FOR COLLEGE STUDENTS THAT HELP REDUCE STRESS AND IMPROVE CREATIVITY.”

OUTPUT:

1. SKETCHING OR DOODLING USING FREE ONLINE TUTORIALS
2. JOURNALING TO IMPROVE SELF-EXPRESSION
3. PHOTOGRAPHY USING A MOBILE PHONE
4. LEARNING GUITAR THROUGH FREE YOUTUBE LESSONS
5. GARDENING WITH SMALL INDOOR PLANTS

ANALYSIS OF THE GENERATED RESPONSES:

1. GENERATING A CREATIVE STORY:

BROAD / UNSTRUCTURED PROMPT RESPONSE

QUALITY:

- SIMPLE AND UNDERSTANDABLE.
- LACKS CREATIVITY AND EMOTIONAL DEPTH.

ACCURACY:

- SINCE STORYTELLING IS IMAGINATIVE, FACTUAL ACCURACY IS NOT IMPORTANT HERE.
- THE STORY IS LOGICALLY CORRECT BUT VERY GENERIC.

DEPTH:

- VERY LIMITED DETAILS.
- NO CHARACTER DEVELOPMENT, SETTING DESCRIPTION, OR CONFLICT EXPLANATION.

BASIC / REFINED PROMPT RESPONSE

QUALITY:

- MORE ENGAGING AND IMAGINATIVE.
- INCLUDES CHARACTER, SETTING, CONFLICT, AND MYSTERY.

ACCURACY:

- INTERNALLY CONSISTENT AND LOGICALLY STRUCTURED.
- MATCHES THE PROMPT REQUIREMENTS CLEARLY.

DEPTH:

- BETTER STORYTELLING DEPTH.
- PROVIDES ATMOSPHERE, PLOT DIRECTION, AND SUSPENSE.

2. ANSWERING A FACTUAL QUESTION:

BROAD / UNSTRUCTURED PROMPT RESPONSE

QUALITY:

- CLEAR BUT EXTREMELY GENERAL.
- USEFUL ONLY FOR BEGINNERS.

ACCURACY:

- FACTUALLY CORRECT.
- HOWEVER, THE INFORMATION IS TOO BROAD TO BE VERY INFORMATIVE.

DEPTH:

- MINIMAL EXPLANATION.
- DOES NOT EXPLAIN TYPES, USES, OR COMPONENTS OF COMPUTERS.

BASIC / REFINED PROMPT RESPONSE

QUALITY:

- WELL-STRUCTURED AND EASY TO UNDERSTAND.
- USES EXAMPLES TO IMPROVE LEARNING.

ACCURACY:

- TECHNICALLY ACCURATE EXPLANATION OF RAM AND ROM.
- CORRECTLY DISTINGUISHES TEMPORARY AND PERMANENT MEMORY.

DEPTH:

- PROVIDES DEFINITIONS, FUNCTIONS, AND EXAMPLES.
- MORE EDUCATIONAL AND PRACTICAL.

3. SUMMARIZING AN ARTICLE OR CONCEPT:

BROAD / UNSTRUCTURED PROMPT RESPONSE

QUALITY:

- CONCISE BUT INCOMPLETE.
- GIVES ONLY THE DEFINITION OF CLIMATE CHANGE.

ACCURACY:

- FACTUALLY CORRECT.
- SCIENTIFICALLY ACCEPTABLE.

DEPTH:

- VERY SHALLOW.
- DOES NOT INCLUDE CAUSES, IMPACTS, OR SOLUTIONS.

BASIC / REFINED PROMPT RESPONSE

QUALITY:

- ORGANIZED INTO BULLET POINTS.
- EASIER FOR STUDENTS TO STUDY AND REMEMBER.

ACCURACY:

- CORRECT CAUSES AND EFFECTS ARE INCLUDED.
- INFORMATION ALIGNS WITH SCIENTIFIC UNDERSTANDING.

DEPTH:

- COVERS MULTIPLE ASPECTS:

- CAUSES,
- ENVIRONMENTAL EFFECTS,
- GLOBAL IMPACT.

4. PROVIDING ADVICE OR RECOMMENDATIONS:

BROAD / UNSTRUCTURED PROMPT RESPONSE

QUALITY:

- ACCEPTABLE BUT GENERIC.
- SUGGESTIONS ARE RANDOM AND NOT TARGETED.

ACCURACY:

- RECOMMENDATIONS ARE VALID HOBBIES.
- NO INCORRECT INFORMATION.

DEPTH:

- NO REASONING OR PERSONALIZATION.
- DOES NOT EXPLAIN BENEFITS.

BASIC / REFINED PROMPT RESPONSE

QUALITY:

- MORE PRACTICAL AND USER-FOCUSED.
- RECOMMENDATIONS MATCH THE USER'S NEEDS.

ACCURACY:

- SUGGESTIONS ARE REALISTIC AND AFFORDABLE FOR STUDENTS.
- BENEFITS ALIGN WITH STRESS REDUCTION AND CREATIVITY IMPROVEMENT.

DEPTH:

- INCLUDES PURPOSE AND CONTEXT.
- MORE THOUGHTFUL AND ACTIONABLE.

CONCLUSION:

REFINED PROMPTS SIGNIFICANTLY IMPROVE AI-GENERATED RESPONSES BECAUSE THEY:

- PROVIDE CLEAR INSTRUCTIONS,
- DEFINE THE CONTEXT,
- SPECIFY AUDIENCE OR PURPOSE,
- AND GUIDE THE AI TOWARD THE EXPECTED OUTPUT.

BROAD PROMPTS OFTEN PRODUCE:

- VAGUE,
- GENERIC,
- AND LESS DETAILED RESPONSES BECAUSE THE AI HAS LITTLE GUIDANCE ABOUT WHAT THE USER TRULY WANTS.